Regime-Aware, Explainable Financial Forecasting Using Multi-Modal Machine Learning

Yashraj Rai

MSc Data Science
Department of Computer Science
The University of Bath
Academic Year 2025–2026

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Submitted by: Yashraj Rai

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Declaration

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Abstract

The financial markets are very complicated systems which include non-stationarity, volatility clusters and shifting behavioural regimes. Forecasting methods tend to become ineffective since they are based on a static relation between past patterns and future values.The dissertation introduces a regime-aware, explainable multi-modal machine learning architecture for forecasting in the field of finance. The architecture includes multiple forecasting experts, such as LSTM, GRU, Transformer and linear models, along with a gating component which is capable of adjusting the choice of the expert depending on the market regime.The architecture includes historical prices, technical indicators, volatility measures and market regime features. Methods of Explainable Artificial Intelligence which uses SHAP is used for analysis of feature importance and model prediction explanation.The proposed solution is compared with traditional forecasting methods and individual machine learning models in terms of forecasting metrics and regime-dependent analysis.

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Acknowledgements

I would like to convey my gratitude to my supervisor, Dr Mingzhi Dong, for all the assistance and support he has continuously provided me in this dissertation work.

In addition to that, I am thankful to the department of computer science at the University of Bath for providing me the resources for completing this dissertation work.

Finally, I am grateful to my family members and friends for their continuous support throughout my MSc degree.

Chapter 1

Introduction

The financial market is a dynamic system impacted by various economic factors, behaviour of the investors, market sentiments, and unexpected events. Predicting the future behaviour of financial assets continues to be one of the most complicated research problems, given the non-stationarity of financial time series, volatility clustering and changes in behaviour over time.

Financial prediction is an attempt to predict future properties of financial assets, including movements in prices, returns, and volatility. The traditional methods of predicting the behaviour of assets have largely been based on mathematical time series models, like the ARIMA model and the GARCH model. Though these models are solid from the mathematical point of view and allow for proper interpretation, they lack robustness when applied to the high-dimensional space and complicated nonlinear dependencies characteristic of the contemporary financial markets.

Machine learning and deep learning have brought new perspectives to financial forecasting. The recurrent network architecture, such as LSTM networks (Hochreiter and Schmidhuber, 1997), helps to identify temporal dependencies in sequential data. The Transformer architecture has enabled modelling the sequences more effectively by applying the attention mechanism (Vaswani et al., 2017).

Nevertheless, the financial forecasting process is difficult since the behaviour of the markets changes over time. Models that are created for particular market conditions might not perform well in other conditions. It implies that models have to account for varying market conditions while forecasting.

1.1 Problem Description

1.1.1 What is financial forecasting?

The term 'financial forecasting' is used to denote prediction of future behaviour of financial assets with respect to price changes, returns, volatilities, etc. Financial forecasting is a significant issue for research in financial analytics since accurate forecasts would help to make decisions regarding investments and portfolio management, for example.

Traditional approaches to financial forecasting use statistical time series models such as the AutoRegressive Integrated Moving Average (ARIMA) and Generalised Autoregressive Conditional Heteroskedasticity (GARCH). The use of these methods is useful for capturing linear dependencies and volatilities in historical data on financial assets. However, financial markets may exhibit non-linear dependencies and operate a large number of information flows, making traditional approaches ineffective for these conditions.

The development of machine learning and deep learning techniques has provided new approaches for financial forecasting by enabling models to learn complex patterns directly from data. Recurrent neural network architectures, such as Long Short-Term Memory (LSTM) networks (Hochreiter and Schmidhuber, 1997), have demonstrated the ability to capture long-term temporal dependencies in sequential financial data. Similarly, Transformer-based architectures introduced through the attention mechanism (Vaswani et al., 2017) have improved sequence modelling by allowing models to focus on important historical patterns without relying solely on sequential processing.

Machine learning and deep learning have helped bring about novel methods of financial forecasting, allowing machine learning models to learn complicated structures directly from the data. One example would be the recurrent neural network models that have shown their ability to learn long-term dependencies of sequential financial data. In addition, architectures like transformers, brought about by the attention mechanism (Vaswani et al., 2017), have contributed to the improvement of sequence modelling through the use of models' ability to focus on important historical trends without the need of sequential processing.

Moreover, financial forecasting is not anymore limited only to the numerical information that can be obtained from the market. Financial news and market sentiment are also rich sources of information which can help predict asset behaviour. Models like FinBERT (Araci, 2019), which are pre-trained language models, allow for the extraction of sentiment information from financial text.

1.1.2 The Exploration Problem in Financial Markets

The financial markets are extremely dynamic systems whose relations between past and future trends vary through time. Unlike conventional forecasting models, the financial time series are non-stationary since their statistical features, like the average value, variance and correlation, do not have to be the same. Moreover, the financial markets tend to show signs of volatility clusters in which volatility is followed by more volatility.

The above-mentioned features present considerable difficulties for the forecasting models. It can happen that a model that was built on the basis of past data under certain market conditions will behave well under similar conditions but will break down under other market conditions. Thus, the patterns of a stable bullish market might become useless during periods of financial instability and market crashes.

Various states of market behaviour are typically described as market regimes. Some examples are bullish, bearish and high volatility regimes, where the dependencies of the financial variables could vary. However, these regimes cannot be observed directly and should be derived from the market data available, such as the return on investment, volatility or the dynamics of the prices. The statistical techniques such as Hidden Markov Models (HMMs) have been extensively used for this purpose (Hamilton, 1989).

The majority of current forecasting methods rely on the assumption that the dependencies of the input variables and the output predictions do not change considerably over time. Yet, such an assumption does not hold in most financial environments because of the possibility of abrupt changes in market behaviour. An important research task is the design of forecasting systems able to recognise the changes in the market conditions and adapt the forecast strategy accordingly.

1.1.3 What is Multi Modal, regime-aware forecasting?

Multi-modal financial forecasting refers to a forecast which integrates the data from multiple data sources to gain a more holistic view of the market behaviour. Instead of relying solely on the history of prices in a market, multi-modal forecasting may include price and volume data, technical data, market sentiment, macroeconomic data, and volatility measurements. Different modalities may complement each other in that financial markets are affected not just by quantitative market movements but also the economic environment.

Nevertheless, the relative significance of the above data sources will depend on the market environment. The features that are significant for making a forecast in the context of a steady bullish environment might have a lesser significance in the context of bearish or highly volatile environment. Regime aware forecasting system extends the concept of multi-modal financial forecasting by taking into consideration the regime of the market while making the forecast.

One possible solution to obtain such a flexible system is by employing Mixture of Experts (MoE) architecture. Mixture of Experts models aggregate several specialised models, called experts, via a gating function, which controls the weight of each expert's contribution to the final output. The adaptive Mixture of Experts architecture, introduced at the beginning of the nineties (Jacobs et al., 1991), showed that different experts may specialise in different parts of an input space, whereas the gating network learns how to aggregate them. More recently, research has been done to extend this idea to finance, with the purpose of developing adaptive trading systems (Ding et al., 2024; Vallarino, 2025).

In the context of financial forecasting, this means that different models may specialise on different temporal patterns or market regimes. Instead of applying a forecasting model to all the observations, the regime-aware gating model assigns different importance weights to experts based on the financial features and regime detected.

1.1.4 Proposed Framework Overview

This thesis presents an approach for a regime-aware and explainable multi-modal forecasting framework that addresses the shortcomings of the current financial forecasting frameworks. This framework uses multiple financial datasets and models in order to be adaptable in any market regime.

The framework uses data from market history, technical indicators, sentiment data, and regimes data. Multiple forecasting experts are integrated within the framework by using an adaptive method such that the framework can change its prediction methodology depending on the market regime. Explainable artificial intelligence methodologies are also used to interpret the results obtained by the models.

The framework architecture, mathematical model, implementation, and evaluation are discussed in detail in the following chapters.

1.1.5 Project Aim

The purpose of this dissertation is to propose and design a regime-aware, explainable multi-modal machine learning model for predicting financial market trends. This research intends to enhance the flexibility of financial forecasting models through the use of a multi-expert system with the help of dynamic selection based on regimes or changing market conditions.

This study will explore the usage of historical market information, technical indicators, sentiment information, and regime-dependent information to predict short-term financial trends using explainable artificial intelligence.

1.1.6 Project Objectives

The objectives of this thesis are:

1. To conduct an extensive literature review and technology review for financial time series forecasting, deep learning methods, financial sentiment analysis, market regimes, and the mixture of experts in architecture and explainable artificial intelligence.
2. To design a regime aware multi-modal forecasting system which uses historical data of the market, technical indicators, sentiment information, and macroeconomic variables by employing multiple forecasting models.
3. To develop a mixture of experts based forecasting system in which the different expert models can exploit different aspects of the behaviour of the financial market and adapt according to the change in market conditions.
4. To study the usefulness of the market regime information for financial forecasting and to see whether the adaptive forecasting system is better performing than the traditional forecasting systems
5. To include explainable artificial intelligence methods like SHAP analysis to find the significant features and to get some insights about the decision - making process of the proposed forecasting system.
6. To benchmark the proposed methodology against the traditional forecasting methodologies and machine learning models using financial forecasting performance evaluation metrics.

1.2 Research Questions

Considering the problems identified in financial forecasting, this thesis addresses the following research questions:

1. Is it possible that the regime aware multi-modal machine learning approach improves the financial forecasting performance in comparison to individual forecasting models?
2. What is the potential of the Mixture of Experts approach for the combination of multiple forecasting models to adjust to different market conditions?
3. Does the use of multiple financial information sources, including historical market data, sentiment data and macroeconomic variables, improve forecasting performance?

4. How can explainable artificial intelligence approaches be used for interpreting the model predictions and determining key factors impacting the financial forecasts?

1.3 Research Contributions

The key contributions of this thesis are:

- Regime-aware multi-modal financial forecasting framework, which integrates multiple financial data, such as market data, sentiment data and regime related data.
- Mixture of experts based forecasting technique that allows for an adaptive combination of different forecasting models based on market conditions.
- Use of explainable artificial intelligence methods for feature importance analysis.
- Experimental validation of the proposed framework by comparing it against baseline forecasting methods based on relevant financial forecasting metrics.

Chapter 2

Literature Review

2.1 Conventional Financial Forecasting and Market Behaviour

Financial forecasting is typically done using conventional statistical methods that attempt to discover patterns and dependencies in historical time series of financial data. This type of approach is built on the idea that there is valuable information in past observations regarding future behaviour of the market and has been extensively used in forecasting prices, returns, and volatilities (Hamilton, 1989; Ozbayoglu, Gudelek and Sezer, 2020).

One of the most frequently used statistical time-series forecasting approaches is Autoregressive Integrated Moving Average (ARIMA). This method allows modelling linear dependencies through autoregression, moving averages, and differencing for the purpose of stationarisation. However, nonlinear market behaviour is common, which makes it difficult to capture all dependencies using pure statistical methods (Ozbayoglu, Gudelek and Sezer, 2020).

Recent advancements in machine learning and deep learning have brought new possibilities such as the development of methods of data-driven learning of temporal patterns based on financial data. Recent advancements include recurrent neural networks, transformers, natural language processing models and multi-modal learning (Hochreiter and Schmidhuber, 1997; Vaswani et al., 2017; Araci, 2019).

Nevertheless, there are still limitations of current forecasting models. Most of the models are based on a single data modality, have static architecture or take into account market stability. These limitations lead to the development of adaptive forecasting models which are capable of combining information from several sources, identifying changing market regimes and providing explainable predictions.

The chapter provides an overview of current literature on traditional financial forecasting, deep learning models, Transformer-based time series forecasting, financial sentiment analysis, market regime identification, Mixture of Experts architectures and explainable artificial intelligence. Limitations of existing approaches are analysed to provide the motivation for the proposed regime-aware multi-modal forecasting model.

2.2 Deep Learning Models for Financial Forecasting

The disadvantages inherent to traditional statistical forecasting models have led researchers to use deep learning techniques in order to model financial time series data. Unlike statistical models, which make use of predefined assumptions about the structure of data, deep learning models are able to learn the complex non-linear structures from big financial datasets.

The most prominent innovation in the field of sequence modelling is the Long Short-Term Memory (LSTM) networks proposed by (Hochreiter and Schmidhuber, 1997). The LSTM network was invented in order to solve the problem of vanishing gradient associated with traditional RNNs, which makes it difficult to capture the dependencies in long sequences.

While the LSTM networks have proved effective in handling temporal relationships, the sequential manner in which the computations are done could be a disadvantage when dealing with large data sets. To mitigate the weaknesses of the networks (Cho et al., 2014),came up with the Gated Recurrent Unit (GRU) architecture. The GRUs are computationally effective while still having the capacity to handle temporal relationships since the recurrent structure of the network is simplified through combining gating processes.

There have also been efforts to incorporate deep learning in probabilistic forecasting. In that regard,(Salinas, Flunkert and Gasthaus, 2017) developed DeepAR, which is an autoregressive recurrent network-based forecasting model that produces probabilistic outputs rather than just point predictions. This is relevant in financial forecasting due to the importance of uncertainties resulting from market volatility and unpredictability of environmental factors.

Another avenue of research has involved continuous time modelling using Neural ODEs.(Rubanova, Chen and Duvenaud, 2018) came up with Neural ODEs as an approach for modelling continuously changing hidden states.

Although there have been significant successes in using deep learning techniques, there are difficulties in using this technique for predicting financial markets. Most of the recurrent neural network models are mostly concerned about past data and may not include other kinds of information sources like financial news, macroeconomic indicators, or different market regimes. In addition, many deep learning models work as single prediction models and hence do not perform well under varying market conditions.

The above drawbacks in using deep learning models have led researchers to investigate attention-based models and multi-modal learning models.

2.3 Forecasting using Transformers

One of the drawbacks of RNNs when considering modelling long-range dependencies is a lack of attention to transformer-based methods for time-series forecasting. Transformers, which were suggested by (Vaswani et al., 2017), allow building self-attentional mechanism-based architectures to discover correlations between positions in a sequence independently of their order. Thus, the usage of Transformer models proved to be promising for modelling complex temporal structures in financial data.

There are some studies applying Transformer-based methods for forecasting problems. Temporal Fusion Transformer (TFT) introduced by (Lim et al., 2020) includes interpretable attention and feature selection for multi-horizon forecasting. Informer (Zhou et al., 2021) is designed to handle the problem of computational complexity of standard Transformer with efficient attention mechanism. Moreover, Autoformer (Wu et al., 2021) and FEDformer (Zhou et al., 2022) introduced long-term forecasting with decomposition and frequency-based techniques. Most recently, (Nie et al., 2023) found out that using patch-based representations might help to increase the quality of long-term forecasting. Nonetheless, the majority of Transformer-based approaches mainly deal with historic numerical data and do not fully solve the problem of handling multiple information sources.

2.4 Language Models in Finance and Sentiment Analysis

Financial markets are driven not only by the numerical movement of prices but also by the information present in financial news, reports, and other related publications (Loughran and McDonald, 2011), showed that financial text requires a special sentiment analysis approach since a general language model might not accurately capture the meaning of financial terms. The rise of Transformer-based language models has greatly increased the efficiency of contextual information extraction in the finance domain.

(Devlin et al., 2019) presented BERT, which made it possible to implement the language context modelling via bidirectional training of the Transformer model. On the basis of this work, Araci (2019) proposed FinBERT – a language model optimised for financial sentiment analysis. Recently, several works have been devoted to the combination of sentiment information and deep learning forecasting models, e.g., explainable stock market prediction framework proposed by (Muhammad et al., 2024). However, sentiment-oriented models are usually built independently of numerical forecasting tools.

2.5 Detection of Market Regimes and Adaptive Forecasting

There exist several regimes that characterise the behaviour of financial markets, namely bullish, bearish, and regimes with high volatility. The regime-switching model was suggested by (Hamilton, 1989) as a tool that helps to represent such systems, where observations are produced from different regimes. Regime-switching became relevant in financial forecasting since models, which were developed for one market state, can function poorly in another market environment.

Several works utilise hidden-state modelling techniques to detect market regimes. For example,(Oelschläger and Adam, 2020) examine Hierarchical hidden markov model to identify the existence of bullish and bearish regimes. Even though market regime detection is helpful in understanding the behaviour of the market, it is often treated as an additional step and not as an integral part of the forecasting framework.

2.6 Multi-modal Financial Forecasting

Multi-modal forecasting refers to the ability to merge data across multiple sources to build a more comprehensive picture of financial activity. Financial prediction could be improved through integration of historical prices, technical indicators, financial text and macroeconomics, as each of these gives a different perspective on market dynamics. TimeGPT-1 by Garza and (Garza and Mergenthaler-Canseco, 2023) showed the potential of foundation models in large-scale time-series forecasting.

Recently, there has been interest in merging multiple financial modalities. CAMEF was introduced by (Zhang et al., 2025), which merges time-series features with macroeconomic announcements, whereas (Tan et al., 2025) explored multimodal learning through mixture-of-experts methods. The difficulty in doing so is that the weight/importance of each modality varies based on the market situation.

2.7 Mixture of Experts and Adaptive Model Selection

Mixture of Experts (MoE) methods allow combining several specialised models using an adaptive gating scheme. The original model for adaptive Mixture of Experts was proposed by (Jacobs et al., 1991), where the gating network is responsible for learning which models should be used in the prediction. Such an approach is especially valuable in solving complicated problems where different models can show better results under various conditions.

Several recent papers have applied expert-based architectures to finance. The paper by (Ding et al., 2024) studied the application of expert models for trading, (Saqur et al., 2024) analysed adaptive gating algorithms for online expert selection.(Vallarino, 2025) proposed a mixture-of-experts approach sensitive to volatility changes for stock forecasting. Nevertheless, the existing methods pay little attention to regime information in the gating scheme.

2.8 Explainable Artificial Intelligence in Financial Forecasting

In recent years, the complexity of machine learning models has led to the demand for explainability in financial forecasting. Despite the predictive power of deep learning models, the interpretability of such models poses difficulties in making decisions in a financial setting.

(Lundberg and Lee, 2017) presented SHAP (SHapley Additive exPlanations), a technique that explains machine learning models' predictions through estimation of feature contributions. (Muhammad et al., 2024) highlighted the significance of applying explainable deep learning for predicting stock market dynamics. Combining explainable techniques in adaptive forecasting can lead to greater transparency of forecasts.

2.9 Research Gap and Overview

There has been tremendous development in financial prediction through the use of statistical models, deep learning architectures, Transformer-based techniques, and financial language models. However, current methodologies tend to address the different aspects of financial prediction separately, without considering a combination of all these different components.

In addition, most of the forecasting techniques have fixed structures and are unable to respond to changes in the economic environment. Even though the mixture of experts algorithms adapts to a certain style through model selection, there is very little work done on the regime aware gating mechanisms in the financial context. This dissertation will seek to fill the gap by developing an explainable regime aware multi-modal forecasting methodology using multiple expert models and multiple data sources for financial predictions.

Chapter 3

Methodology

3.1 Experimental Machine Learning Research Methodology Overview

In this study, a methodology for experimental machine learning research is used in order to explore the potential of a regime-aware explainable multi-modal forecasting framework in the context of financial forecasting.

The methodology encompasses a number of stages such as data acquisition, preprocessing, feature extraction, model development, experiments, and explainability analysis. Multiple financial data sources are combined in order to provide a holistic representation of market behaviour.

First, the historical data from financial markets, sentiments, and economic indicators are acquired. After that, data are preprocessed and converted into useful features that are used as input by the forecasting framework.

The proposed framework is compared against baseline forecasting methods through experiments conducted with the use of quantitative performance metrics. Moreover, in order to assess the impact of different factors, explainable artificial intelligence tools are used.

3.2 Research Design

The research adopts a quantitative experimental research design wherein forecasting models are trained and tested on past financial data with the aim of finding out whether it would be beneficial to combine several sources of information and adaptively select experts in order to outperform traditional methods.

The experiment involves building baseline models and their comparison with the suggested one. Baseline models will include traditional forecasting techniques and individual machine learning algorithms, while the suggested technique will involve combination of several forecasting experts via adaptive selection procedure.

3.3 Description of Dataset

Since financial forecasting demands the combination of several sources of information due to the fact that the market behaviour depends on past price dynamics and other factors, in this research we examine three main types of information: market information, sentiment information and macroeconomic indicators.

3.3.1 Financial Market Data

Past market data is the key component of our forecast model. The database includes OHLCV (Open, High, Low, Close and Volume) data. This set of data provides insight into past market movements and trading volumes.

This information allows us to evaluate past trends in the market and is usually used for financial forecasts. Other market indicators are also calculated on the basis of this data.

3.3.2 Sentiment Data

The financial markets are impacted by sentiments of the investors and information present in the financial news. This is included through the processing of financial text data by means of natural language processing.

The financial language model FinBERT is utilised for generating sentiments from the financial texts. The generated sentiments are then combined with the numeric market features to provide further information on market sentiments.

3.3.3 Macro-economic Variables

Macroeconomic considerations could affect financial market behaviour due to changes in the economic environment. Macroeconomic variables are included as extra inputs to account for macroeconomic considerations.

The inclusion of these variables is meant to capture some information that is not captured by prices alone.

3.4 Data Preprocessing

Financial datasets derived from diverse sources generally involve missing data, format inconsistencies, noise and measurement on different scales. Hence, data preprocessing is a crucial process for preparing the raw data to be used by learning models in order to generate valid forecasts.

The preprocessing steps include data cleaning, alignment of various sources, dealing with missing data, feature scaling, and generation of sequential inputs. This is because the proposed approach integrates market data, sentiments, and macroeconomic factors; thus, all modalities need to be prepared appropriately for the learning processes.

3.4.1 Data Cleaning and Time Series Alignment

Data from financial markets and external sources are gathered at various frequencies. Data related to historical prices may be available daily, while news and economic data may have other publishing frequencies. Thus, all the sources of data are synchronised using a uniform time series index.

The gaps due to missing data are treated using proper methods of data imputation. Any duplication of data is filtered to ensure data quality. The resultant dataset consists of a combined time series that includes market, sentiment, and economic data for each forecasting interval.

3.4.2 Feature Scaling

Differences in the scale of machine learning features affect the performance of the algorithms. For instance, prices, trading volumes, and financial measures differ in scale greatly. Thus, it is important to normalise features prior to the training stage.

This task is performed through standardisation of the features for their transformation into the same scale range while keeping their properties intact. This makes gradient-based optimisation more stable.

3.5 Feature Engineering

The process of feature engineering is done to gain insights into the raw financial data. Financial markets have complicated patterns which might not be apparent from raw price data alone. Thus, extra features have to be generated which reflect market trends, momentum, volatility, and varying market regimes.

The suggested model includes different types of features such as technical indicators based on past prices, volatility features, market sentiment features and market regime features.

3.5.1 Technical Indicators

The technical indicators are those that are computed based on the price-volume data in order to capture the momentum behaviour in the market. Technical indicators like moving average, Relative Strength Index (RSI), Moving Average Convergence Divergence (MACD), and volume trading features are included in the list.

The technical indicators give more information about price behaviour other than just the historical prices.

3.5.2 Volatility Features

Volatility refers to the extent of variation in financial asset prices and can be considered an important indicator in predicting risk and market behaviour. Volatility features are based on historic return series that describe changing market uncertainties.

These features become especially significant when there is a need to identify market instabilities during which prediction algorithms should be changed.

3.5.3 Sentiment Features

Textual sentiment information is used in order to take into account the effect of financial news and people's opinions on the movement of the market.

Textual information of the financial domain is preprocessed using the domain-specific language model in order to extract sentiment features.

3.5.4 Characteristics of Market Regimes

Market regime characteristics are behavioural states of financial markets. These characteristics are derived from the analysis of historical return and volatility patterns for identifying changes in market states.

Information about regimes that is derived through the analysis process is considered as an additional contextual characteristic which allows the framework of forecasting to take into account the state of the market.

3.6 Experimental Methodology

The experimental methodology examines whether the combination of multiple pieces of information sources and forecasting strategy is useful for financial forecasting accuracy.

The experiments involve training the baseline models and then comparing them against the proposed model. The baseline models include classical statistical models and single deep learning models. The proposed model will be tested in order to find out if regime awareness, multi-modal learning and expert selection are better than traditional approaches.

The dataset is split into training, validation and test datasets through time splitting method. This way there is no data leakage involved and the scenario resembles realistic forecasting of finance.

3.7 Performance Evaluation Criteria

Forecasting models are assessed based on several performance evaluation criteria. Because financial forecasting requires the forecasting of both numeric values and movements in certain directions, various criteria are discussed.

3.7.1 Mean Squared Error

Mean Squared Error (MSE) is a metric used to measure how well a model's predictions match the actual values. It calculates the average of the squared differences between predicted and actual values.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

where y_i represents actual value and $\hat{y}_i$ represents predicted value

3.7.2 Mean Absolute Error

Mean Absolute Error (MAE) is calculated as the sum of absolute errors divided by the sample size.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$$

A high MAE could mean the model needs to be retrained or adjusted. A lower MAE gives more confidence that the model is producing useful estimates.

3.7.3 Directional Accuracy

In the context of financial forecasting, the importance of accurate prediction of the price change direction can be noted. Directional accuracy reflects the percentage of correct forecasts on price change direction made by the model.

$$DA = \frac{Correct\ Direction}{Total\ Predictions} \times 100$$

This criterion is especially valuable for analysis of trading-oriented forecasting models.

3.7.4 Explainability Evaluation

SHAP-based feature attribution analysis is used to evaluate explainability, where SHAP values are analysed in order to identify the contributions of different features to predictions by the models.

The goal is to analyse how feature importance changes in different market regimes and if the decisions regarding forecasting are still interpretable.

3.8 Summary

In this chapter, the methodology of building and evaluating the proposed forecasting framework was provided. The methodology covers data collection and preprocessing, feature engineering, and evaluation.

Different sources of financial information are processed and converted into machine learning inputs. The next chapter is dedicated to the proposed multi-modal regime-aware forecasting framework.

Chapter 4

Proposed Framework

4.1 Framework Overview

To address the limitations of current financial forecasting approaches, this thesis aims at developing an explanatory, regime-aware, multi-modal machine learning system. The problem with traditional financial forecasting models is that they use only one source of information and operate under the assumption that historical dependencies remain unchanged over time. On the other hand, financial markets are always changing depending on the economic situation and investors' activity.

The model incorporates a number of financial perspectives and forecasters. There is a dynamic gating system that allocates weights based on the market regime identified.

The system architecture comprises four modules: multi-modal input representation, market regime recognition, forecasting experts and adaptive experts .Fusion of the input representations takes place, followed by passing on these representations to the regime recognition and gating layers that dynamically weigh forecasting experts. A consensus decision is obtained by combining expert decisions. Dynamic weights are assigned to forecasting experts by the gate network. The final prediction is computed based on the predictions of forecasting experts. Explainable artificial intelligence techniques are used for analysing feature importance and explaining the model decisions in various market regimes.

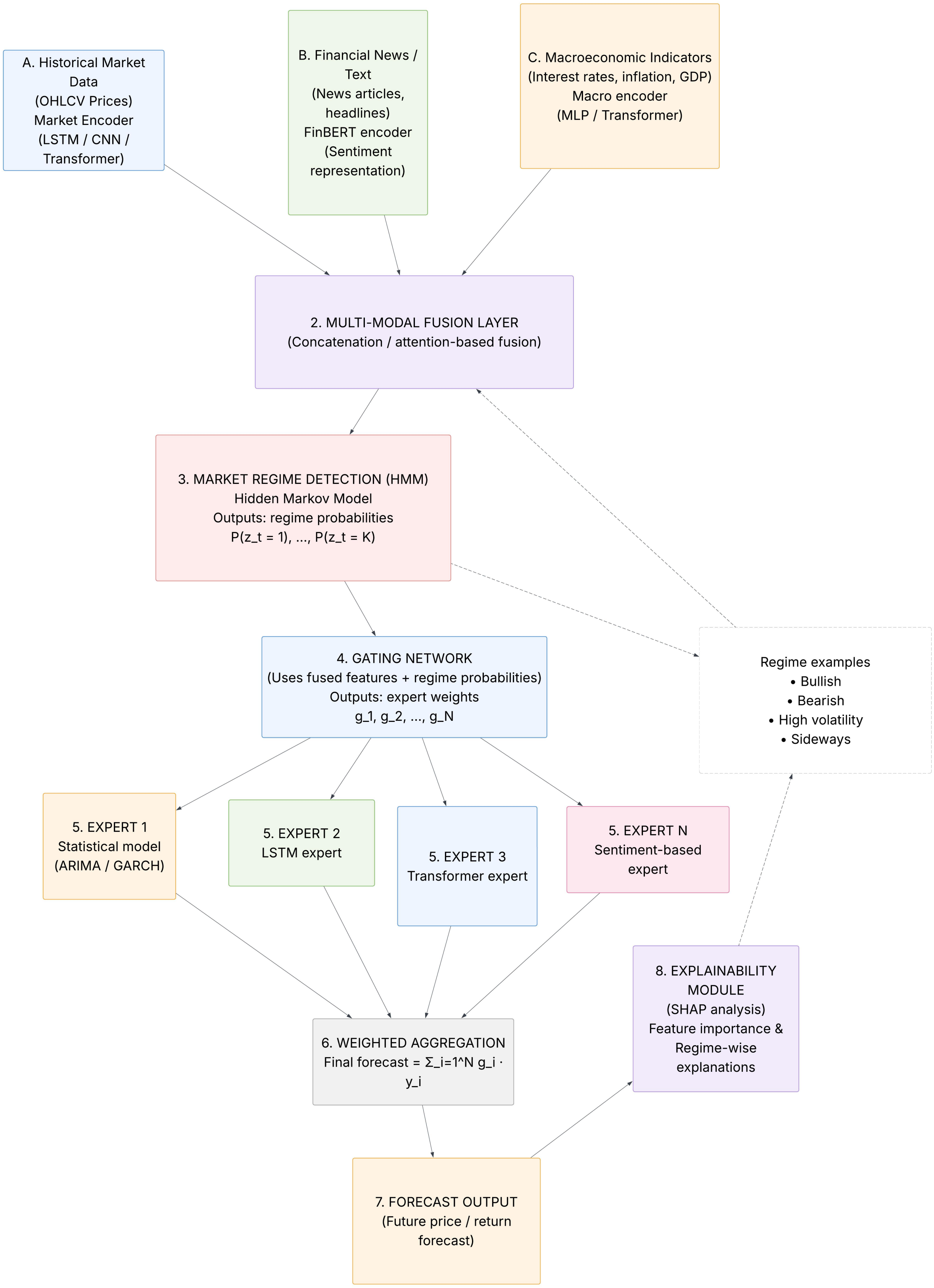

Figure 4.1: Proposed regime-aware multi-modal financial forecasting architecture

4.2 Multi-Modal Input Representation

The performance of financial markets depends on many factors, and thus depending on the history of price changes may not give enough data to conduct an accurate forecast. Therefore, the suggested framework uses three input modalities – numerical market data, financial sentiment data, and macroeconomic indicators.

Each type of data is represented separately by means of appropriate representation techniques and then combined with a fusion approach. This way, each information source retains its unique features while adding extra value to the process of forecasting.

4.2.1 Representation of Financial Market Data

The past market data serves as the major source of data used in financial forecasting. The OHLCV data is used to represent the price movement and trading behaviour.

The financial time-series data is represented as:

$$X_p = \{x_1, x_2, ..., x_t\} \tag{4.1}$$

Where each observation is a set of market data points at a specific time step.

Several technical indicators are derived using historical market data to represent the trends, momentum, and volatility in the market data. This representation provides additional insights other than the raw price value and makes the model capable of identifying the temporal patterns.

The processed market data is fed into the sequence modelling algorithms such as LSTMs and transformers.

4.2.2 Sentiment Representation

The financial market is influenced by information included in news articles, financial statements, and the views of investors. Thus, the suggested framework considers textual sentiment information to use it for forecasting.

Financial text is represented numerically using a language model designed specifically for the financial domain. FinBERT is chosen because it is trained specifically on financial documents.

The sentiment representation is described by the following equation:

$$X_s = FinBert(text) \tag{4.2}$$

where X_s is the sentiment features extracted from the financial text data

These sentiment features represent information about the events taking place outside the market and expectations of the investors that are not included in the historical price information.

4.2.3 Representation of Macroeconomic Features

There are several ways through which the economy affects the behaviour of financial markets using factors like interest rate, inflation, and economic growth, among other factors. Hence, macroeconomic features will be introduced as another form of input data.

The macroeconomic features vector is denoted by:

$$X_m = \{m_1, m_2, ..., m_n\} \tag{4.3}$$

Where each variable represents an economic feature. Using macroeconomic features gives an opportunity to consider wider economic factors instead of relying only on asset history.

4.3 Market Regime Identification Module

Financial markets do not exhibit steady behaviour over time. Different periods, like bull markets, bear markets and high volatility periods, signify different regimes where forecasting models might have varying performance.

In order to solve this problem, the proposed architecture incorporates the market regime identification module. The function of the module is to identify hidden regimes in the market and provide extra context to the forecasting system.

Hidden Markov Models (HMMs) are used due to their capability to model systems where financial behaviour is driven by hidden states.

4.3.1 Regime Detection Using a Hidden Markov Model

The hidden Markov model postulates that financial data is produced by some hidden market states. The hidden states are those which describe the various behavioural states of the financial market.

The hidden regime state variable is denoted as:

$$z_t = \in \{z_1, z_2, z_3\} \tag{4.4}$$

where:

- z_1 represents a bullish market condition
- z_2 represents a bearish market condition
- z_3 represents high volatility conditions

The posterior distribution of the present regime state is defined as follows:

$$P(z_t | x_{1:t}) \tag{4.5}$$

Here, z_t denotes the regime state, and $x_{1:t}$ denotes historical observation

4.3.2 Integration of Regime Features

The information about the regimes obtained from the above analysis is integrated into the multi-modal representation of features. The regime detection process does not occur as a stand-alone module, but regime information is used for forecasting directly.

The representation is formed as:

$$h = [X_p, X_s, X_m, X_r] \tag{4.6}$$

where:

- x_p represents market features.
- x_s represents sentiment features
- x_m represents macroeconomic features
- X_r represents regime information

4.4 Multiple Expert Forecasting Models

In the suggested approach, multiple experts are used rather than relying on one prediction model. These experts specialise in detecting various patterns found in the financial data.

There are various forecasting approaches that have unique properties. While statistical approaches work well in case of linear dependencies, deep learning algorithms work well for nonlinear dependencies.

4.4.1 Statistical Forecasting Expert (ARIMA/GARCH)

A statistical forecasting expert is considered a traditional forecasting approach in finance. The ARIMA forecasting model is used to capture linear dependencies in the financial time series, while the GARCH forecasting model is applied to capture volatility changes.

The reason for adding statistical models to the forecasting scheme is to have an easily interpretable forecasting model able to capture established financial dependencies.

Nevertheless, the drawback of statistical models is that they cannot capture nonlinear dependencies and can utilise only one source of information.

4.4.2 LSTM Forecasting Expert

The Long Short Term Memory (LSTM) network is a type of recurrent neural network capable of discovering long-term dependencies from sequential data.

The LSTM forecasting expert has been added since there exist temporal dependencies in financial time series which could span several observations. By virtue of their architecture, LSTM networks are able to store useful information from history while ignoring the useless one.

In our proposed model, the LSTM expert is concerned with learning the nonlinearities from market histories.

4.4.3 Transformers Forecasting Expert

The Transformer expert was added to account for long dependency structures in the financial time series data. As opposed to recurrent models, Transformer models employ the attention mechanism to learn the connections between observations in a sequence.

Attention allows for the assigning of weights to the historical observations according to their importance. This ability can be helpful in financial forecasts, as previous events may affect future behaviour.

The Transformer expert acts as a supplementary forecasting expert compared to recurrent neural network-based experts.

4.4.4 Sentiment-Based Forecasting Expert

The sentiment-based forecasting expert deals with modelling the relationship between financial text information and future market behaviour.

The sentiment expert obtains representations for sentiment that are produced by financial language models and learns about the impact of changes in market sentiment on assets.

In contrast to numerical forecasting experts, which mainly deal with analysis of price patterns in the past, the sentiment expert obtains information from outside the market.

The addition of this expert allows us to incorporate financial information into our framework along with market information.

4.5 Multi-Modal Feature Fusion

Different views of the financial market behaviour are captured through different information sources. Price data has a temporal perspective, sentiment data reflects the influence of financial market events, while macroeconomic indicators capture a more broad-based economic view of the problem. Nonetheless, the different information sources possess different characteristics and structural features. Thus, a good mechanism for integrating the different information sources is needed.

The architecture incorporates a feature fusion layer where the extracted representations of the various modalities are fused. The purpose of this fusion layer is to generate a unified view of the market, which incorporates all possible information sources.

The fused representation is illustrated as:

$$H = [H_p, H_s, H_m] \tag{4.7}$$

where:

- H_p refers to the historical market features encoding.
- H_s refers to the sentiment encoding.
- H_m refers to the macroeconomic encoding.

And the fused encoding is then mapped to a shared feature space using a neural network layer

$$H_f = F(H) \tag{4.8}$$

Here, H_f is the final representation obtained after considering the context. The fusion enables the forecasting model to incorporate the connection between the sources of information as opposed to analysing them separately.

4.6 Regime-Aware Mixture of Experts Structure

A key feature of the suggested model is the use of a regime-aware Mixture of Experts (MoE). Most of the existing forecasting models typically apply one model for any market situation. Nevertheless, financial markets behave differently at different times, which means that some forecasting approaches can be suboptimal in some regimes.

The suggested model overcomes this drawback by considering multiple forecasting experts and a regime-aware gate. The gate network learns to weigh the experts differently based on the current regime and the multi-modal representation.

The final forecast is calculated as a weighted average of expert predictions:

$$\hat{y} = \sum_{k_1}^{K} g_k f_k(x) \tag{4.9}$$

where:

- $\hat{y}$ represents the final prediction value
- $f_k(x)$ represents the prediction generated by an expert k
- g_k represents the weight assigned to the expert k
- k represents the total number of experts This proposed design allows the framework to adapt automatically to the forecasting strategy

4.6.1 Gating Network

The gating network is the module that selects which expert should contribute. Rather than manually specifying which forecasting model to use under certain market regimes, the gating network learns which experts to combine based on historical data.

The input to the gating network is the fused multi-modal representation plus the market regime:

The final output is produced using a weighted combination of the experts:

$$G = softmax(W_g H_f + b_g) \tag{4.10}$$

where:

- w_g represents the gating network weights

- b_g represents the bias term
- G represents the probability distribution over forecasting experts

It is because of the softmax function that the expert weights generated become a probability distribution where the sum of the weights of all experts is equal to one.

For instance, in the case of a stable regime in the market, more emphasis could be placed on the statistical models, while during the volatile times, the deep learning experts could be given more weightage.

4.6.2 Expert Selection Algorithm

The expert selection algorithm is responsible for the role of forecasting models in the formation of the resulting forecast. Experts are specialised in extracting different types of relations from financial data.

The statistical expert is responsible for the extraction of linear relationships and volatilities. The LSTM expert extracts sequential relations, while the Transformer expert extracts long-term relations with the use of the attention mechanism. Finally, the sentiment-based expert takes into account information from financial texts.

The gating network dynamically weights the contribution of experts depending on:

- Current market regime.
- History of market performance.
- Sentiment conditions.
- Macroeconomic environment.

4.6.3 Final Forecast Prediction

The final prediction is formed by aggregating all predictions of experts using the weights provided by the gating network.

The prediction generation process can be formulated as follows:

$$\hat{y} = g_1 f_1(x) + g_2 f_2(x) + ... + g_K f_K(x) \tag{4.11}$$

Such aggregation allows making use of the strengths of various models while minimising the reliance on one particular method.

The predicted value will reflect the future behaviour of the asset based on past trends, sentiments, the macroeconomic environment, and the current market regime.

4.7 Objective for Training and Optimisation

This proposed methodology involves joint optimisation of forecasting experts, gating mechanism, and regime encoding. The objective function is formed in such a way that ensures accurate forecast results as well as proper expert selection.

Optimisation objective function is formulated as:

$$L = L_{forecast} + \lambda_1 L_{regime} + \lambda_2 L_{diversity} \tag{4.12}$$

where:

- $L_{forecast}$ represents forecasting error.
- L_{regime} encourages regime-aware expert selection
- $L_{diversity}$ prevents experts from learning similar behaviours
- λ_1 and λ_2 control the contribution of additional losses.

4.7.1 Forecasting Loss

The forecasting loss function gives us the measure of discrepancy between real and estimated values.

The mean square error is taken as the criterion for forecasting:

$$L_{forecast} = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{4.13}$$

Minimisation of this loss function leads to obtaining more accurate forecasts.

4.7.2 Regime Consistency Loss

The regime consistency loss term makes sure that the gating network generates weights of the experts which are consistent with the regime discovered from the markets.

This makes sure that forecasting experts are selected in accordance with the regime prevailing in the markets.

4.7.3 Expert Diversity Loss

One limitation that a mixture of Experts faces is expert collapse, where different experts learn very similar representations.

Diversity loss makes sure that the experts exhibit different predictive behaviours. This will allow the experts to specialise in different parts of financial prediction.

4.8 Explainability Framework

Models for financial forecasting are viewed as hard to interpret because of their complex nonlinear structure. However, interpretability is essential in financial applications since some decisions can be life-changing.

The suggested framework uses Explainable Artificial Intelligence (XAI) with SHAP values for analysing the forecasted results and features' importance.

The process of explainability is executed after the prediction is generated in order to discover which features play an important role in the forecasts.

4.8.1 Feature Attribution using SHAP

SHAP (Shapley Additive Explanations) is employed to quantify the contribution of individual features to model predictions.

The prediction explanation is presented as:

$$f(x) = \phi_0 + \sum_{i=1}^{n} \phi_i \tag{4.14}$$

where:

- ϕ_i represents the contribution of the feature i
- ϕ_0 represents the baseline prediction

SHAP analysis gives insight into how price direction, volatility, sentiment, and macroeconomic features contribute to predictions.

4.8.2 Explanation Regime Analysis

This framework expands on SHAP analysis through regime-wise analysis of feature importance.

Feature importance might vary based on market regimes. For instance, volatility-based features might have a higher importance in high-risk situations, while sentiment-based information might be more important at critical times in the market.

Regime-wise SHAP analysis gives us insight into how the forecasting framework adjusts its decision-making process in light of changing market regimes.

4.9 Summary

In this chapter, the proposed framework for regime-aware and explainable multi-modal financial forecasting has been introduced. This framework considers the historical data of the markets, financial sentiments, and macroeconomic factors via multi-modal fusion techniques.

A Hidden Markov Model helps in identifying the regimes in the market, whereas a Mixture of Experts model helps in the adaptive selection of the forecasting model based on the market regimes.

A SHAP-based explainability approach has been introduced in order to understand the significance of the features. In the following chapter, the implementation and experiments of the proposed framework are discussed.

Chapter 5

Implementation

5.1 Implementation Overview

This chapter provides an overview of the implementation procedure for the proposed regime-aware, explainable multi-modal financial forecasting system. After the proposed design of the framework as presented in Chapter 4, this chapter discusses the development environment, data processing procedure, model implementation and experimental setup.

The implementation procedure includes various steps, such as data collection, data preprocessing, feature generation, implementation of individual forecasting modules, and Mixture of Experts architecture along with an explainability procedure.

The entire framework was implemented in the form of various machine learning and deep learning modules based on Python.

5.2 Implementation of Data Processing Pipeline

The data processing pipeline transforms unstructured financial information into structured inputs that can be fed to machine learning models.

As the proposed architecture involves many modalities, every input data source needs individual processing before integration.

5.2.1 Data Collection

Historical stock market data was collected from financial data providers that had OHLCV data. Financial text data was collected to create sentiments out of it, whereas macroeconomic data was collected to understand economic scenarios.

All the data sets were synchronised based on their time stamps to create a financial forecasting data set.

5.2.2 Implementation of Data Preprocessing

The preprocessing was carried out to ensure that the financial data gathered could be used during training. This involved data cleansing, data normalisation, and sequential sampling.

The resulting dataset was then split into training, validation, and test datasets based on time to avoid future data leakages.

5.2.3 Feature Engineering Implementation

Feature engineering was done to get useful features out of financial data.

Technical features like moving averages, RSI, MACD, and volatility were calculated based on past prices.

Sentiment features were extracted from financial text data using the financial language model representation.

The final list of features included the following:

- Historical price features.
- Technical features.
- Volatility.
- Sentiment.
- Macroeconomic features.
- Market regime features.

5.3 Implementation of Forecasting Models

The presented approach includes several forecasting experts. All of them were implemented individually and then included in the Mixture of Experts structure.

5.3.1 ARIMA/GARCH Expert Implementation

A linear expert was used as an elementary forecasting model in the Mixture of Experts scheme. It offers a basic forecasting viewpoint and introduces diversity into the system along with the deep learning experts.

5.3.2 LSTM Model Implementation

LSTM expert was constructed by making use of sequential financial data to identify temporal dependency within the historical observations. The forecast was used as one of the expert models in the Mixture of Experts setting.

5.3.3 Implementation of Transformer Model

Transformer Expert was implemented using the self-attention technique to learn the long-range dependencies in the financial sequence. The forecasting from the transformer expert was combined with others through the gating mechanism.

5.3.4 Implementation of Sentiment Model

This GRU expert is used to model sequences because it has a gated structure. It can serve as an added capacity for forecasting sequences within the expert network.

5.4 Implementation of Regime-aware Mixture of Experts Model

The final model was developed using the combination of individual forecasting experts and the market regime detection system and the adaptive gating system.

Each expert generates its own forecast independently, and the gating network decides the weight of each expert based on the prevailing market condition. This enables the model to adapt its forecasting strategy in response to the prevailing financial environment.

5.4.1 Regime Detection Based on HMM

A hidden Markov model was employed to detect various regimes in financial time series data sets.

The algorithm uses historical data in terms of return and volatility features to learn hidden regimes in the data set, which can be used as an additional context for the gating network.

The process involves:

- Calculation of historical return from price data.
- Creation of volatility-based features.
- Training the Hidden Markov Model on historical observations.
- Estimation of probability distribution of the current market regime.
- Provision of regime information to the expert selection process.

It should be noted that the output of the hidden Markov model is not just one regime, but a probability distribution of the regimes. This helps to deal with uncertainty in transition between regimes.

5.4.2 Implementation of the Gating Network

The gating network is a component of the neural network designed to assign dynamic weights to the forecasting experts.

The input of the gating network includes the fused multimodal representation along with the market regime identification results.

The output of the gating network is the probability distribution of the set of experts based on the softmax activation function. The generated weights reflect the importance of each expert in producing the prediction.

Implementation steps are as follows:

1. Retrieve fused feature representation.

2. Merge the representation with regime data.
3. Forward pass the merged representation through fully-connected layers.
4. Softmax activation to get expert weights.
5. Weigh experts' predictions using the obtained probabilities.

This way, the proposed method can automatically determine the best forecasting approach rather than using one universal model for all market regimes.

5.4.3 Combining Experts and Prediction

Once all the forecasts from all the forecasting experts, as well as the weights for them from the gating network, have been acquired, the final forecast will be generated via a weighted combination method.

The algorithm works in accordance with the Mixture of Experts method, which uses the final prediction calculated based on the importance of each expert.

The forecast pipeline involves the following steps:

- Generating separate forecasts from each expert.
- Calculating expert weights with the help of the gating network.
- Multiplying the predictions of each expert by their respective weight.
- Combining the weighted predictions to create the final output.

This technique enables specialised forecasting models to make contributions based on their strengths depending on the prevailing market situation.

5.5 Implementation of Explainability

Explainability is an important part of the proposed framework because financial forecasts need to be transparent and understandable.

The explainability module was developed using SHAP (SHapley Additive explanations). It allows one to analyse the feature-level explanations of model predictions.

The following questions were answered in the course of the SHAP analysis:

- What is the importance of various input features?
- How much does each variable contribute to predictions?
- How feature importance changes in various market regimes?

The explainability analysis allows one to understand whether the financial forecasting model uses important financial signals or learns meaningless patterns.

5.5.1 SHAP Integration

Integration of SHAP to the forecasting process helps to study the connections between the input features and the output predictions.

The trained forecasting algorithm serves as an input to the SHAP explainability method that calculates the contributions of individual features to the output prediction.

The integration process consists of the following steps:

1. Selection of trained forecasting models.
2. Provision of test instances to the SHAP explainability method.
3. Calculation of contribution values of features.
4. Visualisation of the global and local importance of features.

The explanations are utilised for the analysis in Chapter 6.

5.5.2 Analysis of Regime-Wise Explanation

Along with the analysis of feature importance in general, SHAP explanation is also carried out independently for each market regime.

This type of analysis enables the study of how the importance of financial features changes depending on market conditions.

For instance, features related to volatility can acquire higher importance during periods of high market volatility, while sentiment features can gain higher importance during critical market events.

Therefore, regime-wise analysis is an additional proof that the suggested model adjusts its decision process according to market conditions.

5.6 Experimental Setup

An experimental setup is required to describe the environment in which the training and testing of the forecasting framework takes place.

The splitting process of the dataset for training, validation, and testing sets was carried out by using time-based splitting in order to make sure that no future data is used for training of models which is realistic in a financial forecasting case.

The following comparisons were made through experiments:

- Traditional statistical forecasting methods.
- Deep learning forecasting methods.
- Transformer-based forecasting methods.
- Proposed regime-aware multi-modal mixture of experts model.

5.6.1 Training Process

Training Process involves successive optimisation of the forecast model architecture.

The steps involved are:

1. Processing and preparation of multi-modal inputs.

2. Training the forecast experts.
3. Training the gating network based on expert outputs and regimes.
4. Optimisation of the overall forecast model objective function.
5. Evaluation of the performance of the model on validation data.

The training process is continued until satisfactory performance is achieved on the validation data set.

5.6.2 Configuration of Hyperparameters

It is very important to use proper hyperparameters in order to achieve optimal machine learning model performance. The key hyperparameters are:

- Learning rate.
- Batch size.
- Number of training epochs.
- Number of hidden layers.
- Number of expert models.
- Length of sequence in case of time-series data.

The hyperparameters have been chosen experimentally based on validation results.

5.7 Summary

This chapter has explained the development of the proposed regime-aware explainable multi-modal financial forecasting framework.

The development included creation of various forecasting experts, regime detection using Hidden Markov Model, Mixture of Experts framework and SHAP explainability approach.

The designed pipeline allows the framework to use multiple financial information sources, adapt to market dynamics and make explainable forecasting decisions.

The following chapter discusses the experimental results of the proposed framework compared with other forecasting methods.

Chapter 6

Results and Evaluation

6.1 Overview of Experimental Results

This chapter will discuss the experimental analysis of the regime-aware Mixture of Experts framework for forecasting. The analysis will cover the aspects of forecast performance, learning dynamics, market regime classification, expert selection adaptability and explanations.

The regime-aware Mixture of Experts consists of four experts - LSTM, Transformer, GRU and linear forecasters. The output of the experts is combined by the gating network that uses the information about market regime from the Hidden Markov Model.

The evaluation of the model is conducted on unseen samples through MAE, RMSE and directional accuracy metrics. Also, the regime-aware Mixture of Experts approach is compared to baseline models such as ARIMA, LSTM and the equally weighted expert ensemble to measure the benefit of adaptive expert selection.

Also, the analysis of features importance and decision-making processes of both gating mechanism and forecasting experts through SHAP is conducted.

6.2 Training and Convergence Behaviour Study

The training performance of the proposed regime-aware Mixture of Experts model was assessed by studying the changes in the training and validation scores over 30 epochs. Convergence behaviour was studied using the cumulative loss from the forecast and Mean Absolute Error (MAE).

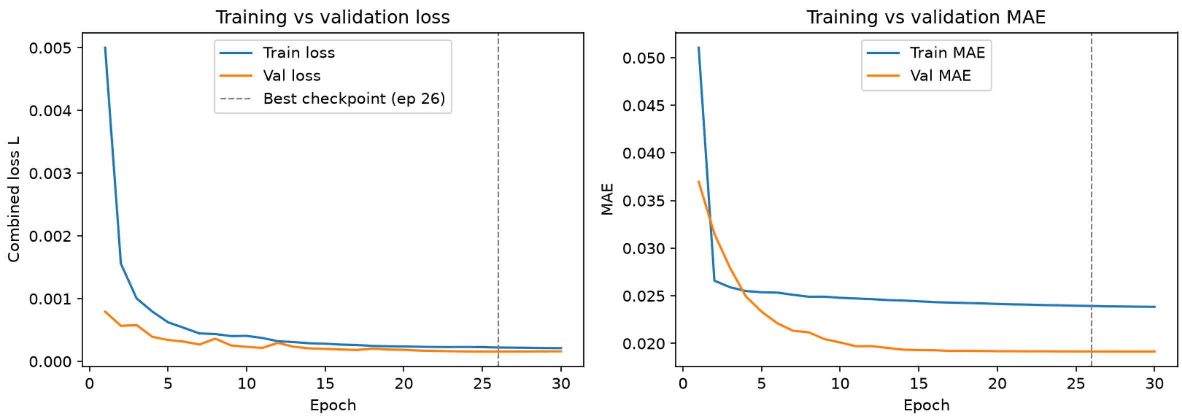

Figure 6.1: Training and validation loss and MAE during model optimisation

It is worth noting that the model performs well during the training phase, with the training loss reducing from 0.005000 to 0.000211 from the first to last epochs, respectively. The same pattern of reduction applies for the validation loss which is recorded at 0.000793 and 0.000160 in the first and last epoch respectively.

In addition, the validation MAE decreases consistently throughout the training period from 0.036960 at the first epoch to approximately 0.019152 at the last epoch of the training process, indicating that the model learns the required patterns within the financial time-series dataset.

The best performance on the validation set is observed at epoch 26, when the model records a validation loss of 0.000158 and a validation MAE of 0.019157. At this point, the model's validation performance becomes stable, indicating that there is limited improvement beyond this point.

The training and validation performance curves are quite similar, and an increase in validation error is not observed during later epochs, implying that there is minimal overfitting.

6.3 Forecasting Performance Evaluation

Performance of forecasting of the proposed regime-based mixture of experts framework was evaluated based on the test data set that has not been used before. This was done based on three metrics: Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and Directional Accuracy (DA).

Test data set comprised 28,836 sequences built from 11 input variables. Proposed model has four forecasting experts: Long Short-Term Memory (LSTM), Transformer, Gated Recurrent Unit (GRU), and Linear forecasting expert. Output of these experts is mixed using gating network, which was learned based on market regimes detected using the Hidden Markov Model.

Goal of this experiment is to check the possibility to produce accurate forecasts using adaptive approach.

6.3.1 Performance of Proposed Framework Test

The trained model was tested on the independent test set with the help of the best checkpoint generated during training (epoch 26).

The proposed framework performed with the following results:

Metric	Value
MAE	0.02228
RMSE	0.03178
Directional Accuracy	58.20%
Forecast Loss	0.0002
Regime Loss	0.0011
Diversity Loss	0.0000

Table 6.1: Test performance of the proposed architecture

The developed forecasting method produced MAE of 0.02228 and RMSE of 0.03178 on the test dataset. Directional accuracy of 58.20% means that the developed forecasting approach has correctly predicted the direction of future movement for more than half of the test observations.

Along with the above mentioned performance indicators, the percentage based errors have also been computed. The developed forecasting approach produced MAPE of 124.11% and SMAPE of 153.84%. Though these percentages may seem quite high, however, they are not used as performance indicators for primary research. The goal of the forecasting is to predict financial returns and actual values of such returns are often very close to zero. As MAPE and SMAPE are computed using the actual value, the percentage errors are very high due to low actual values.

Thus, MAE, RMSE and directional accuracy should be used as performance indicators of the forecasting methodology.

6.3.2 Comparison with Baselines

To assess the performance of the regime-aware Mixture of Experts framework, the model was compared with the three baseline methods such as ARIMA, Single LSTM and the mean value of the forecasting experts.

ARIMA was used as the traditional statistics-based forecasting method and Single LSTM for the contribution of using multiple forecasting experts. An ensemble average method was considered to compare the learned gating method with equal weightage of all the experts.

All the models were tested on the same test set with 28,836 samples. MAE, RMSE, and DA were used as the evaluation metrics.

Model	MAE	RMSE	Directional Accuracy
ARIMA	0.02271	0.03204	51.68%
Ensemble Average	0.02410	0.03336	51.81%
Single LSTM	0.02228	0.03175	58.36%
Proposed MoE Framework	0.02228	0.03178	58.20%

Table 6.2: Comparison between proposed framework and baseline models

The ARIMA baseline obtained a directional accuracy of 51.68%, showing poor performance with regards to capturing the complicated dynamics existing in financial time series. While the ARIMA baseline got comparable MAE with a value of 0.02271, it showed less efficiency with regard to directional prediction compared with the deep learning models.

The unweighted ensemble method provided the same directional accuracy of 51.81% but showed greater prediction errors than the proposed method. The results suggest that a simple average of several expert forecasts cannot offer the same benefits as the learning of dynamic expert contributions using the gating network.

The single LSTM baseline had the highest directional accuracy among all individual methods at 58.36%, along with the highest RMSE of 0.03175. The proposed MoE framework showed almost similar performance with a directional accuracy of 58.20% and RMSE of 0.03178.

While the proposed method does not show any better numerical prediction performance compared with the Single LSTM baseline, the main advantage of the proposed framework lies in its adaptive expert selection procedure. Contrary to the single LSTM framework, the proposed framework adjusts the contribution of different experts depending on the market regime detected.

6.4 Analysis of Market Regime

In the suggested model, the Hidden Markov Model (HMM) is used to provide information about the market regime to the gating network. The posterior probabilities of the HMM are used as additional input features that help to recognise different market regimes when making predictions.

In the suggested model, three regimes were considered: bullish regime, bearish regime, and high volatility regime. The dominant regime of each test sample is found by choosing the regime with the maximum posterior probability.

The distribution of the identified regimes on the test set is shown in Table 6.4.

Regime	Samples	Percentage
Bull	20246	70.21%
Bear	1110	3.85%
High-volatility	7480	25.94%

Table 6.3: Distribution of detected market regimes

According to the findings, there is not a single stationary behaviour within the test period but various market behaviours. A considerable amount of data falls under the category of high volatility regimes, demonstrating the relevance of adaptive forecast techniques. This section examines the impact of regime signals on expert selection.

6.5 Gating Mechanism and Expert Behaviours

The goal of the Mixture of Experts architecture is to integrate various forecasting models based on the prevailing market situation. Instead of using an unweighted combination of experts, the proposed model applies the gating network to weigh contributions of each expert differently.

In this section, the behaviour of the gating mechanism will be examined in order to see whether the model adjusts the choice of experts to the specific market environment.

6.5.1 Expert Weight Adaptation in Different Market Regimes

The gate network produces a distribution of weights among the four forecasting experts: LSTM, Transformer, GRU, and linear models.

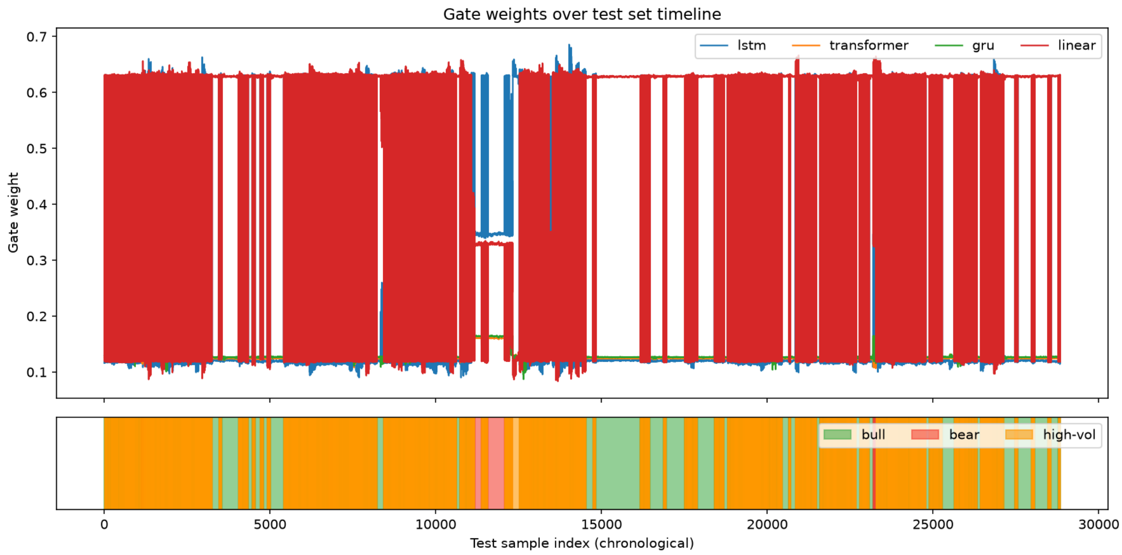

Figure 6.2: Expert gate weights across test samples with detected market regimes

These results indicate that the gating network does not provide stable expert contributions through the entire testing phase. It appears that the role of the experts is constantly modified in accordance with the prevailing market regime.

In the case of a rising market regime, the linear expert receives the maximum contribution value of 0.630. However, during high volatility regimes, the LSTM expert becomes more prominent and provides a contribution of 0.629.

Finally, during the bearish regime, the gating network gives a higher contribution to the LSTM expert of 0.346, and still provides contributions from the other experts.

6.5.2 Expert Selection Behaviour

Regime	LSTM	Transformer	GRU	Linear
Bull	0.118	0.125	0.127	0.630
Bear	0.346	0.161	0.164	0.330
High-volatility	0.629	0.126	0.125	0.120

Table 6.4: Average expert weights under different market regimes

From the differences between the expert weights, we see that the gating network learns to select distinct forecasting methods in different regimes.

Linear expert outperforms in the bullish regime, showing that a simple relationship is enough for a predictable regime. In the case of high-volatility regimes, the LSTM expert gets the highest weight from the gating network, implying that sequential approaches work better for complex regimes.

The standard deviation of the gate weights across regimes is 0.1133, which means that the expert contributions differ in a significant way across regimes.

6.6 Explainability Analysis

The explainability is a vital part of the suggested framework since it involves multiple forecasting experts whose contributions vary based on the state of the market. The SHAP (SHapley Additive exPlanations) method was used for the analysis of the gating network and forecasting experts in order to determine the feature impact on them.

6.6.1 Analysis of Gating Mechanisms using SHAP

The gating network controls how each expert contributes in predicting the value based on dynamic weights. The SHAP analysis was done on the gates to get insight about which features affect the expert selection process.

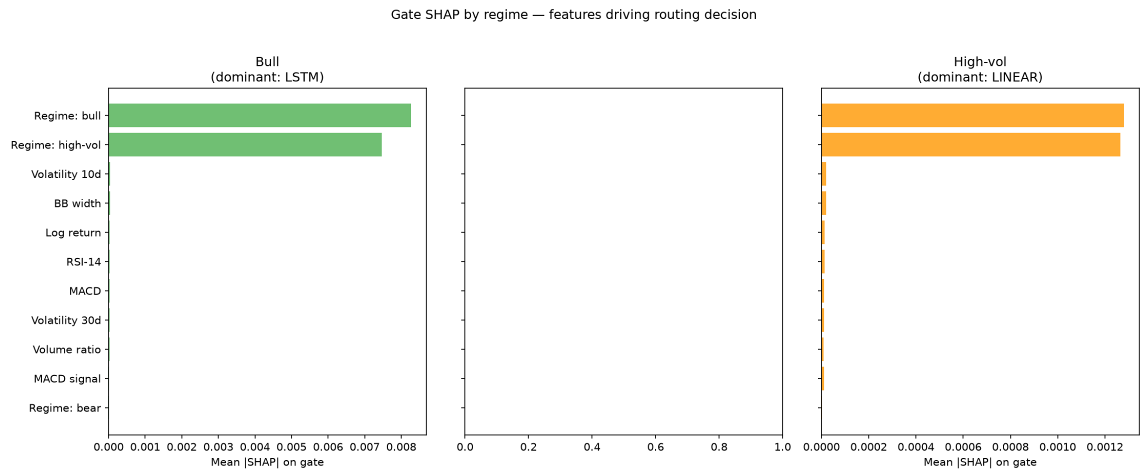

Figure 6.3: SHAP feature importance influencing gating decisions under different market regimes

It is clearly seen from the results that the information about regimes has considerable impact on routing in the gating network. In the case of bullish environment, regime encoding has considerable impact on the selection of experts, and the high-volatility regime also has considerable influence on routing process.

It fits the purpose of the architecture in question when the gating network modifies the contribution of experts depending on the current market environment instead of assigning fixed weights to the experts.

At the same time, SHAP values for the bearish regime could not be computed reliably enough. It is explained by the comparatively low number of bearish observations as compared to bullish and high-volatility regimes.

6.6.2 SHAP Analysis of Forecasting Experts' Predictions

Besides analysing the gating network, SHAP analysis was also conducted on the forecasting experts to determine the features that influence their predictions.

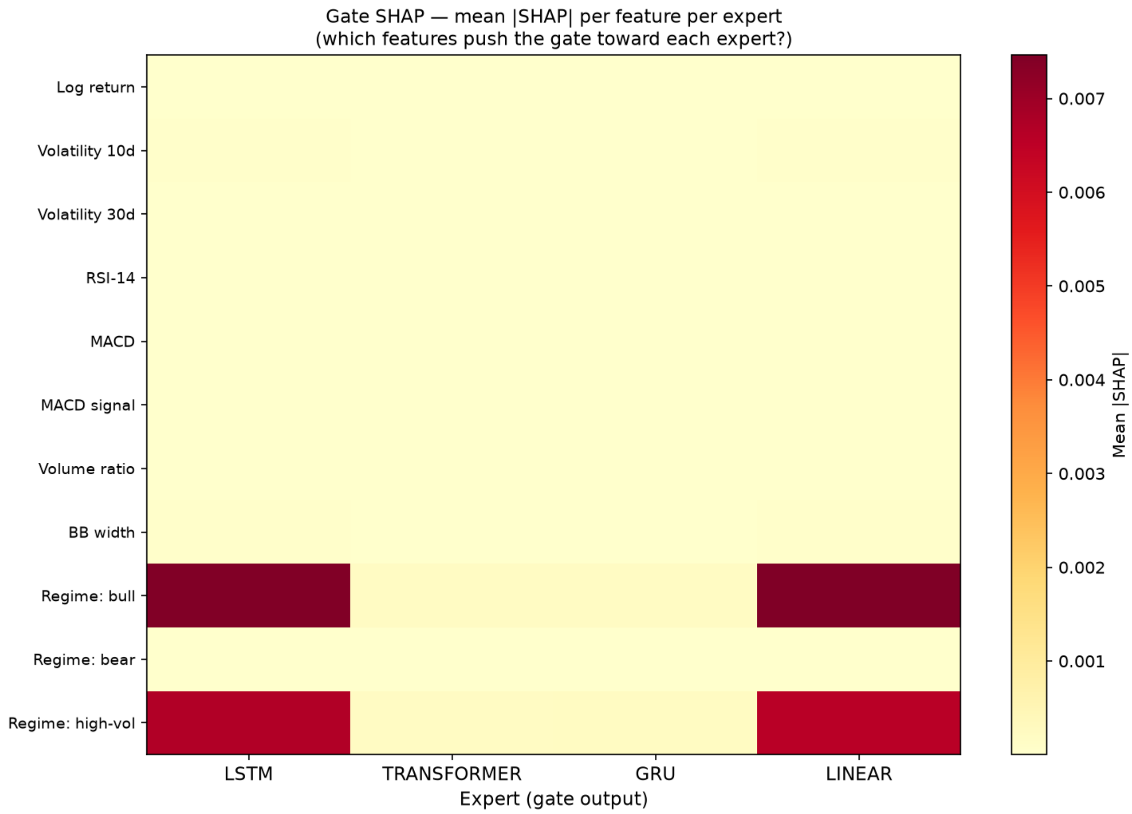

Figure 6.4: Feature importance influencing individual expert predictions

These findings show that experts have different financial features on which they base their forecasts. In turn, this shows the rationale for the Mixture of Experts method, where experts' models can take into account different relationships existing in the financial market.

Sequential experts, such as LSTM and GRU, are capable of capturing temporal relationships based on market behaviour in the past, while the Transformer expert captures more generalised relationships among the input features. The linear expert acts as an additional simple forecasting tool, which is especially important in the case of market stability.

Thus, the SHAP analysis additionally confirms that the developed model does not act as a single black-box forecasting model. Different experts contribute through specialised behaviour, while the gating network determines their significance depending on market conditions.

6.7 Limitations of Experimental Evaluation

Despite the presented adaptive Mixture of Experts model's ability for adaptive expert selection and competitive forecasting results, there were several limitations revealed.

The framework was able to provide comparable forecasting performance with Single LSTM baseline framework in terms of error metrics rather than providing significant improvements.

Thus, the major contribution of the framework is in its adaptive expert selection and regime awareness.

The efficiency of the gating mechanism relies on the accuracy of the market regime detection. Besides that, due to the low number of bearish observations, the regime-specific analysis became less reliable and bearish SHAP analysis lost its value.

While the framework is capable of utilizing different sources of financial data, the role of sentiment and macroeconomic variables individually should be analyzed further.

Chapter 7

Discussion

7.1 Key Findings

In this chapter, we present the results that have been achieved using the regime-aware multi-modal mixture of experts approach. In this study, the aim was to determine whether an adaptive approach can offer comparable financial predictions while being able to adapt to market changes better and deliver understandable model decisions.

The experiments show that the presented framework manages to combine several forecasting experts using regime-aware gating. Even though this model was not able to show any significant improvements over the single LSTM baseline in terms of traditional forecasting measures, it was shown that the model can adapt the weight of each expert based on the current market conditions. Thus, the main contribution of the framework is its adaptive decision-making capabilities.

7.2 Forecasting Performance Discussion

When compared to the performance of baseline models, the proposed model was shown to be comparable to the performance of the independent LSTM model while surpassing the performance of conventional statistical forecasting and equally weighted ensemble forecasting models in directional forecast performance.

It is clear from the limited improvements in numerical forecasting metrics that the advantages of the MoE approach lie in model flexibility. Financial markets are affected by varying conditions and thus one single model cannot always represent all aspects of the market behaviour. This problem is overcome in the proposed model as different experts have different weight contributions depending on different market conditions.

7.3 Adaptive Expert Weighting and Financial Regimes Identification

The results of expert weights analysis prove that the gating network learns how to allocate experts based on financial regimes. The linear expert was contributing the most in bullish periods, while the LSTM expert played a more important role when volatility was higher. This means that there is no need for the fixed importance of experts and it is possible to adapt the forecasting strategy depending on the market characteristics.

The results justify the idea of using regime information as another context for financial forecasts. Based on the information from the Hidden Markov Model, the gating structure can determine when one architecture of forecasting is more appropriate.

Chapter 8

Conclusion

The development of an explainable machine learning framework, which is aware of the financial market regime, was explored in this dissertation. Financial markets are dynamic systems, and their performance depends on many factors, such as the economic situation, investor behaviour and some unexpected occurrences, which makes financial forecasting difficult. In traditional methods of forecasting, constant relationships are assumed between elements of financial time series, and only single forecasting experts are used, thus limiting the ability of forecasting models to adapt to changes in the market. In order to solve these problems

The suggested approach is a combination of various forecasting methodologies using the framework of the Mixture of Experts model, where each expert specialises in learning certain aspects of the financial data. The Hidden Markov Model was applied for recognising various regimes and providing the context for the gating network. Depending on the recognised regime, the gating network adapts the weight of each individual expert, such as LSTM, GRU, Transformer, and Linear models. Therefore, the approach can go beyond a constant forecasting strategy and behave depending on the market environment.

The empirical investigation revealed that the suggested framework is capable of performing competitive forecast accuracy on the unseen test data. In particular, the framework provided a directional accuracy of 58.20% without sacrificing an equivalent error performance to the benchmark deep learning models. While the suggested framework did not outperform the single LSTM model in terms of traditional forecasting accuracy metrics, the investigation highlighted the importance of an adaptive selection of the most appropriate expert. The expert contribution analysis revealed that various forecasting models were more prominent in different regimes of the financial market. The Linear expert was more contributive during the bullish market regime, while the LSTM expert performed better during high volatility periods.

One of the key contributions of the work presented here lies in the incorporation of the concept of explainability in the framework of forecasting. Financial forecasting models have often been found to be lacking in this aspect, especially in the context of employing deep learning techniques. The usage of SHAP analysis helped us understand the factors that were affecting the expert selection process and individual experts' forecasts.

The results of this dissertation will aid in building adaptive forecasting systems for finance using regime detection, multiple forecasting methods, and explainable decision-making into one framework. This dissertation has shown that in order to improve forecasting systems, one does not need to focus only on making prediction error smaller but also on other factors such as adaptiveness and interpretability.

Although the following limitations should be considered, several achievements have been made. First, the discovered market regimes were not balanced enough, having fewer bearish observations for certain regime-based analysis. Second, although the approach allows for using several financial data sources, it is still necessary to conduct an additional investigation to estimate the impact of each feature separately. Also, future research could focus on larger multi-market datasets, better regime discovery techniques, additional forecasting experts, and more ablation studies.

Number of words until this point, excluding front matter: 10793.

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Appendix A

Software Developement Tools

A.1 Development Environment

The suggested framework was developed through a Python-based machine learning environment. Python was chosen due to the vast ecosystem for finance data analysis and explainable artificial intelligence.

The development environment contains libraries for numerical computations, data processing, statistical modelling, deep learning, and model explanation.

A.1.1 Programming Language and Libraries

The programming language chosen for the development of the forecasting model is Python.

The primary libraries used are:

- **Pandas and NumPy**: For data manipulation, numerical operations and time series operations.
- **Scikit-learn**: For data preprocessing, statistics and machine learning operations.
- **PyTorch/TensorFlow**: For building deep learning models (LSTM and Transformer).
- **Statsmodels**: For statistical forecasting models such as ARIMA.
- **HMM libraries**: For market regimes identification.
- **SHAP**: For explanation and feature attribution.

A.1.2 Hardware and Software Configuration

All the experiments were carried out using a machine learning environment that comprises of Python, computation libraries and hardware necessary for training deep learning algorithms.

The software configuration was as follows:

Table A.1: Environment of Implementation

Component	Configuration
Programming language	Python
Deep Learning Library	PyTorch/TensorFlow
Data processing library	Pandas, NumPy
Algorithm explanation library	SHAP

Appendix B

Results from Additional Training and Evaluation

This is an additional set of experimental results that have been generated as part of the training and testing phases for the proposed regime-aware Mixture of Experts framework.

Table B.1 presents the main configuration details of the training process.

B.1 Training configuration summary

Table B.1: Training configuration summary

Parameter	Value
Training epochs	30
Input features	11
Training sequences	134,678
Validation sequences	28,836
Test sequences	28,836
Number of forecasting experts	4
Experts	LSTM, Transformer, GRU, Linear
Model parameters	966,440
Best checkpoint epoch	26

B.2 Training Progress

Table B.2 shows selected checkpoints from the training procedure, along with training loss, validation loss, validation MAE, and direction accuracy.

Table B.2: Selected training progress checkpoints

Epoch	Train Loss	Validation Loss	Validation MAE	Validation DA
1	0.0050	0.0008	0.03696	0.426
5	0.0006	0.0003	0.02334	0.463
10	0.0004	0.0002	0.02010	0.522
15	0.0003	0.0002	0.01931	0.559
20	0.0002	0.0002	0.01918	0.577
26	0.0002	0.0002	0.01916	0.583
30	0.0002	0.0002	0.01915	0.584

B.3 Final Test Evaluation Output

Table B.3 shows the evaluation metrics obtained during the final testing on unseen data.

Table B.3: Final test evaluation results

Metric	Value
MAE	0.02228
RMSE	0.03178
Directional Accuracy	58.20%
MAPE	124.11%
SMAPE	153.84%
Forecast Loss	0.0002
Regime Loss	0.0011
Diversity Loss	0.0000